E-COMMERCE PLATFORM FOR PERSONALIZED RETAIL EXPERIENCES: PERSONALIZED PROMOTION ENGINE

25-26J-526

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Dissertation submitted in partial fulfillment of the requirements for the Bachelor of Science (Hons) in Information Technology Specializing Software Engineering

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DECLARATION

I declare that this is my own work and this dissertation1 does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

In today's highly competitive retail sector, supermarkets worldwide are increasingly adopting personalization to enhance customer engagement and boost sales. Traditional promotional campaigns are generic and broadcast to all customers, often leading to low click-through rates (CTR), poor conversion, and significant wasted marketing expenditure. This research report presents the design, implementation, and evaluation of an AI-Based Personalized Promotion Engine, developed as a core component of a larger project: an E-Commerce Platform for Personalized Retail Experiences

The proposed engine analyzes customer purchase history, behavioral patterns, and loyalty data to generate individualized promotions including tailored discounts, product bundles, cross-sell incentives, and seasonal offers in real time. The system integrates several machine learning paradigms like Collaborative Filtering (CF) for user-item similarity modeling, Content-Based Filtering (CBF) for product attribute matching, K-Means clustering for RFM-based customer segmentation, and a Random Forest classifier for purchase probability prediction. Additionally, uplift modeling principles are applied to estimate the incremental effect of each promotional treatment on individual purchasing behavior, ensuring that promotions are allocated where they will produce the greatest business impact.

The engine was developed and evaluated using publicly available datasets the Instacart Market Basket Analysis dataset, the UCI Online Retail dataset, Web scraped Keells supermarket dataset supplemented with synthetic retail data generation to simulate realistic promotional scenarios. The system was integrated via RESTful APIs into web (React) and mobile (React Native) front-end applications.

Evaluation results demonstrate substantial improvements over traditional mass-promotion approaches. The personalized engine achieved a Return on Investment (ROI) of 566.7%, a purchase conversion rate of 25%, and a marketing cost reduction of 75% compared to broadcast campaigns. Precision@50 was recorded at 14%, with the top-ranked customer cohort exhibiting 100% purchase probability scores. Clustering quality, assessed by Silhouette score, confirmed well-defined customer

segments. These results validate that an AI-driven, uplift-aware personalization approach significantly outperforms generic promotional strategies in both customer engagement and business value.

The research contributes to the growing body of literature on promotion-aware recommender systems by uniquely combining uplift modeling with supermarket-specific business rules, real-time API delivery, and customer segmentation in a single integrated pipeline an approach currently absent in both commercial platforms and academic prototypes.

Keywords: Personalized Promotion Engine, E-Commerce, Machine Learning, Recommender Systems, Uplift Modeling, Customer Segmentation, Collaborative Filtering, RFM Analysis, Supermarket Retail, Artificial Intelligence

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Special appreciation goes to the academic community whose open-source contributions made this research possible particularly the contributors of the Instacart Market Basket Analysis dataset, the UCI Machine Learning Repository, and the open-source developers behind TensorFlow, PyTorch, scikit-learn, and related frameworks.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
CBF	Content-Based Filtering
CF	Collaborative Filtering
CTR	Click-Through rate
DB	Database
RFM	Recency, Frequency, Monetary (customer segmentation)
XAI	Explainable Artificial Intelligence

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1. INTRODUCTION

The retail industry is undergoing rapid transformation driven by advances in digital technology and the growing expectation of personalized consumer experiences. In the supermarket sector, where competition is intense and customer loyalty is fragile, the ability to deliver relevant, timely, and individualized promotional offers has emerged as a critical strategic differentiator. However, despite significant investment in digital platforms, many supermarkets continue to rely on generic, broadcast-style promotional campaigns that treat all customers as a homogeneous group. This one-size-fits-all approach leads to wasted marketing resources, low customer engagement, and missed revenue opportunities.

The proliferation of e-commerce has generated unprecedented volumes of customer behavioral data including purchase histories, browsing patterns, product interactions, and loyalty card records. These data streams, when analyzed through advanced machine learning techniques, provide the foundation for building highly accurate, personalized promotional systems. International retail giants such as Amazon, Walmart, and Tesco have demonstrated that AI-driven personalization can significantly increase conversion rates, customer retention, and overall revenue. However, such enterprise-grade solutions are typically expensive, proprietary, and tailored to large-scale operations, leaving small and mid-tier supermarkets without accessible, effective alternatives.

This research addresses this gap by designing, implementing, and evaluating an AI-Based Personalized Promotion Engine specifically tailored for supermarket e-commerce contexts. The engine integrates multiple machine learning paradigms Collaborative Filtering (CF), Content-Based Filtering (CBF), K-Means clustering, Random Forest classification, and uplift modeling within a unified promotion generation pipeline. Unlike existing recommender systems that focus solely on product ranking, this engine goes further by selecting the optimal promotional treatment for each individual customer whether a percentage discount, a product bundle, a cross-sell incentive, or a seasonal offer based on predicted behavioral response and business constraints.

This report represents the individual research contribution of Nayanapriya H T E (IT22367044) toward the group project titled 'E-Commerce Platform for Personalized Retail Experiences' (Project ID: 25-26J-526), supervised by Prof. Dilshan De Silva. The group project encompasses four integrated components: a Sinhala Voice Assistant, the Personalized Promotion Engine (this component), a Business Intelligence Dashboard with Explainable AI, and a Dynamic Customer Segmentation and Loyalty System. This report focuses exclusively on the Personalized Promotion Engine.

1.1 Background

Supermarket retailing is a high-volume, low-margin industry where customer loyalty is paramount. Globally, supermarkets spend significant portions of their revenue on promotional activities including discounts, coupons, loyalty schemes, and seasonal campaigns. Despite this investment, the effectiveness of traditional promotional approaches is increasingly questioned. Research consistently shows that generic promotions delivered to entire customer bases produce significantly lower engagement than targeted, personalized alternatives.

The limitations of traditional promotion systems are well-documented. Broadcast campaigns suffer from low click-through rates (CTR), typically in the range of 1-5%, meaning that the vast majority of promotional spend is directed at customers who have no interest in the promoted product or offer. This inefficiency drives up the cost per acquisition, reduces return on investment (ROI), and contributes to customer fatigue a phenomenon where customers begin to ignore promotions due to their irrelevance.

The emergence of machine learning and big data analytics has created new possibilities for promotion personalization. By analyzing patterns in transaction data what customers buy, how often, when, and in what combinations it becomes possible to predict individual customer preferences with high accuracy. Recommendation systems, which have been widely deployed in streaming services, e-commerce, and social media, provide a foundational technology for this purpose. However, their

application to the specific challenge of promotion treatment selection determining not just what to recommend, but which type of incentive will most effectively motivate a purchase remains relatively underexplored.

Uplift modeling, a technique from the field of causal inference and treatment effect estimation, offers a promising approach to this challenge. Unlike traditional predictive models that estimate the probability of a purchase occurring, uplift models estimate the incremental effect of a promotional intervention the change in purchase probability attributable specifically to receiving the promotion. This distinction is crucial: a customer who would have made a purchase regardless of receiving a promotion represents wasted promotional spend, while a customer who is on the boundary of a purchase decision and can be tipped by the right offer represents the highest-value target for promotional resources.

The research presented in this report builds upon these foundations to develop a practical, deployable system that brings together the best elements of recommendation systems, customer segmentation, and uplift modeling within a coherent, API-integrated promotion engine. The system is designed to be lightweight, customizable, and accessible to small and mid-tier supermarkets markets currently underserved by enterprise solutions.

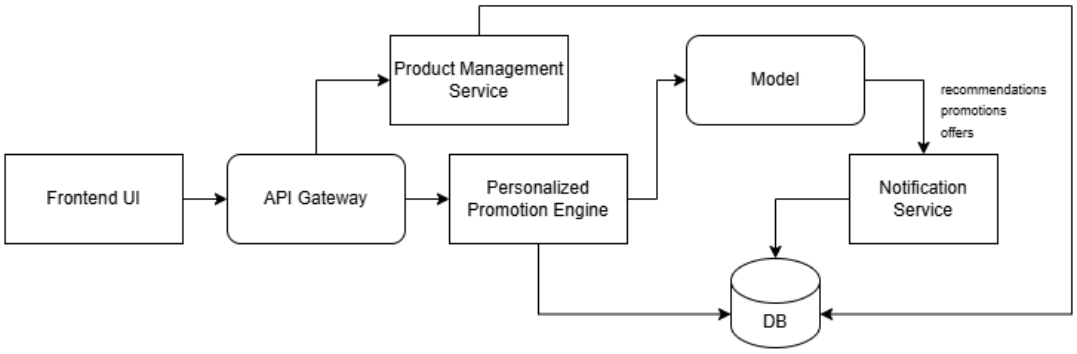

Figure 1.1: System Overview Diagram of the Personalized Promotion Engine

1.2 Literature Survey

1.2.1 Recommender Systems in E-Commerce

Recommender systems have been a subject of intensive research since the 1990s, with significant milestones including the development of collaborative filtering techniques. The foundational paradigms of CF and CBF have been extensively studied and refined, with hybrid approaches combining both methods proving most effective in practice. CF operates by identifying users with similar purchase histories and recommending items favored by those similar users. CBF, by contrast, analyzes the attributes of items a user has interacted with and recommends similar items. Both approaches have demonstrated strong performance in product recommendation tasks across e-commerce, streaming, and social platforms.

Deep learning has recently extended the capabilities of recommender systems significantly provided a comprehensive survey of deep learning-based recommender systems, demonstrating that neural collaborative filtering, recurrent neural networks (RNNs), and attention mechanisms can capture complex, non-linear user-item interactions that elude traditional matrix factorization methods. These advances have enabled more accurate modeling of sequential purchase behavior and temporal patterns in customer data capabilities that are particularly valuable in supermarket contexts where weekly shopping patterns and seasonal trends are prominent features of customer behavior.

A comprehensive review of deep learning applications in e-commerce personalization was conducted in [1], identifying that state-of-the-art systems increasingly combine multi-modal data (text, images, behavioral logs) with sophisticated model architectures to achieve highly granular personalization. However, the review also noted that many of these advanced systems are computationally demanding and require large datasets, limiting their applicability in smaller retail contexts.

1.2.2 Promotion-Aware Recommendation Systems

While item recommendation has been extensively studied, promotion-aware recommendation where the goal is not just to suggest products but to determine the optimal promotional treatment has received comparatively less attention. Personalized promotion in e-commerce was first explicitly addressed in [2], where a framework was proposed to jointly model user preferences and promotion responses. Their work demonstrated that incorporating promotion sensitivity into recommendation models could significantly improve promotional effectiveness compared to standard collaborative filtering approaches.

This line of work was extended to consider item-ranking optimization in the presence of promotional interventions [3], demonstrating that promotion-aware ranking models could outperform generic recommendation systems in terms of both engagement and revenue generation. Hybrid approaches combining collaborative filtering with association rule mining for coupon promotion were explored in [4], demonstrating the value of cross-selling logic in promotion targeting. Their results showed that customers receiving bundle recommendations based on their co-purchase patterns had significantly higher basket values than those receiving standard single-item promotions.

A dynamic adaptive model (DAMA) that accounts for promotion influence in recommendation was proposed in [5], showing that models which explicitly incorporate promotional effects as features outperform those that treat promotions as external interventions. This finding underscores the importance of treating promotion type as an integral component of the recommendation decision, rather than an afterthought applied post-recommendation.

1.2.3 Customer Segmentation in Retail

Customer segmentation is a foundational technique in retail marketing, enabling businesses to differentiate their approach to distinct customer groups based on behavioral, demographic, or value-based criteria. The RFM (Recency, Frequency, Monetary) framework is one of the most widely applied segmentation approaches in

retail, offering a simple but powerful way to classify customers based on how recently they made a purchase, how often they purchase, and how much they spend [6].

K-Means clustering is the most commonly applied algorithm for RFM-based segmentation, owing to its computational efficiency and interpretable results. K-means clustering was applied to fashion retail customer data in [7], identifying distinct segments including high-value loyalists, at-risk churners, and occasional buyers, each requiring different promotional strategies. Clustering approaches in recommender systems were reviewed more broadly in [8], noting that segmentation can provide valuable prior information to recommendation models, improving cold-start performance for new users.

More advanced segmentation approaches include hierarchical clustering, DBSCAN, and Gaussian Mixture Models, each offering different trade-offs between interpretability, computational complexity, and robustness to outliers. For the present research, K-Means was selected as the primary segmentation algorithm due to its suitability for the RFM feature space and its compatibility with downstream promotion allocation logic.

1.2.4 Uplift Modeling and Causal Inference

Uplift modeling, also known as incremental modeling or true-lift modeling, addresses the causal question of whether a promotional treatment will causally increase a customer's purchase probability, rather than merely predicting which customers are most likely to purchase regardless of receiving a promotion. The distinction is crucial: a model that targets only the most likely buyers regardless of treatment effect will waste promotional budget on customers who would have converted without any incentive [9].

Causal forests and meta-learners (S-learner, T-learner, X-learner) are among the most commonly applied methods in uplift modeling. The effectiveness of combining causal forests with reinforcement learning for dynamic marketing uplift modeling was demonstrated in [9], achieving significant improvements in incremental conversion rates over baseline approaches. Their work is directly relevant to the

present research, as it establishes the theoretical and empirical foundations for treatment-effect-aware promotion targeting.

Despite these advances, commercial implementation of uplift modeling in supermarket retail remains limited, primarily due to the data requirements for treatment-control experimental designs and the computational complexity of causal inference methods. The present research addresses this gap by implementing a simplified but practical uplift-aware promotion selection framework that is tractable for deployment in real-world supermarket contexts.

1.2.5 Association Rule Mining for Bundle Generation

Association rule mining, pioneered with the Apriori algorithm [10], identifies co-purchase patterns in transaction data that can be leveraged to generate product bundle recommendations. The technique produces rules of the form "if a customer buys product A, they are likely to also buy product B," quantified by support, confidence, and lift metrics.

In the context of supermarket promotion, association rules provide a computationally efficient and interpretable method for identifying cross-selling opportunities. Customers who purchase bread can be targeted with milk discounts; customers who buy coffee can be offered complementary promotions on creamer or sugar. This approach has been widely validated in grocery retail contexts [4] and forms a core component of the bundle promotion strategy in the present engine.

1.3 Research Gap

Despite the extensive literature on recommender systems, customer segmentation, and uplift modeling, a significant gap exists in the practical integration of these approaches within a single, real-time, API-driven promotion engine designed for supermarket e-commerce contexts. Several specific gaps can be identified:

- Existing recommender systems focus on product ranking rather than promotion treatment selection. Most academic and commercial systems predict which products a customer will prefer, but do not extend this to determine which type

of promotional incentive discount, bundle, cross-sell, or seasonal offer will be most effective for that customer at that moment.

- Commercial personalization platforms lack uplift modeling. Platforms such as Amazon Personalize and Adobe Target are highly effective at item ranking but do not incorporate causal uplift estimation, meaning they cannot distinguish between customers who will buy regardless of a promotion and those who genuinely require a promotional nudge to convert.
- Business rule integration is absent from academic models. Research prototypes for promotion personalization typically operate in controlled experimental settings and do not address the integration of real-world business constraints such as promotion budget limits, fairness requirements across customer segments, and inventory-driven promotional triggers.
- Supermarket-specific contextual factors are underrepresented. Most existing systems are developed for general e-commerce contexts and do not account for supermarket-specific factors such as seasonality, local cultural preferences, weekly shopping cadences, and perishable product constraints.
- Lack of real-time API integration. Academic models are predominantly offline systems that cannot deliver personalized promotions at the individual level in real time through web and mobile interfaces.

Table 1.1: Comparison of Existing Personalization Systems vs. Proposed Solution

Capability	Amazon Personalize	Adobe Target	Academic Models (Hybrid CF, CBF, Clustering)	Proposed Promotion Engine
Item Recommendations (ranking)	Yes	Yes	Yes	Yes
Personalized Promotions (discounts, bundles)	No	Limited (rule-based)	Basic, not optimized	Fully automated and data-driven
Uplift Modeling (incremental sales)	No	No	No	Core novelty
Budget & Business Rule Constraints	No	Manual setup	Absent	Integrated

Real-Time Adaptation (API)	Yes	Yes	Offline only	Yes
Supermarket-Specific Context (seasonal, cultural factors)	No	No	No	Explicitly included
Commercial Deployment Readiness	Mature SaaS	Mature SaaS	Experimental	Lightweight & customizable

This research directly addresses these gaps by proposing a novel integration of CF, CBF, K-Means clustering, Random Forest classification, uplift modeling principles, and association rule mining within a unified real-time promotion engine. The proposed system is designed to be deployable in supermarket e-commerce contexts, with explicit support for business rules, budget constraints, and seasonal factors capabilities absent from both commercial platforms and existing academic prototypes.

1.4 Research Problem

Supermarket e-commerce platforms continue to rely heavily on generic promotional campaigns that are distributed uniformly to all customers, irrespective of individual purchasing behavior, preferences, or responsiveness to promotional incentives. This approach results in significant marketing waste, reduced profitability, and poor customer engagement. Even where basic personalization has been adopted, existing systems predict what a customer might like but do not determine which promotional treatment a discount, bundle, a cross-selling incentive, or a seasonal offer will produce the greatest incremental effect on that customer's purchasing decision.

This problem is compounded in supermarket contexts by the complexity of customer behavior: shopping patterns vary by day of week, season, cultural occasion, and life stage; price sensitivity differs across customer segments; and the interaction effects between products mean that optimal promotions are often bundles rather than single-item discounts. Existing systems lack the integration of causal uplift estimation with

real-time business rule enforcement that would be required to address these complexities.

Research Problem: How can an AI-driven promotion engine generate real-time, individualized promotional treatments that maximize incremental sales uplift, while balancing business rules, budget constraints, fairness, and transparency in a supermarket e-commerce context?

1.5 Research Objectives

1.5.1 Main Objective

To develop an AI-based personalized promotion engine that dynamically recommends promotional treatments based on individual customer behavior and preferences, achieving measurable improvements in click-through rates, purchase conversion rates, and return on investment compared to generic mass-promotion approaches.

1.5.2 Specific Objectives

1. Analyze customer purchase histories and interaction logs to extract behavioral insights including RFM features, purchase frequency, and product affinity patterns.
2. Build and evaluate machine learning models, including Collaborative Filtering, Content-Based Filtering, K-Means clustering, and Random Forest classification, for customer segmentation and purchase probability prediction.
3. Develop an uplift-aware promotion treatment selection mechanism that prioritizes customers most likely to respond incrementally to promotional interventions.
4. Generate tailored promotions including percentage discounts, product bundles, cross-sell incentives, and seasonal offers using an integrated promotion generation pipeline combining ML outputs with business rules.
5. Integrate the personalized promotion engine with web and mobile interfaces through RESTful APIs, enabling real-time promotion delivery.

6. Evaluate the engine's performance using CTR, purchase conversion rate, Precision@K, ROI, and clustering quality metrics, benchmarking against traditional generic promotion approaches.
7. Assess the commercialization potential of the system as a deployable solution for small and mid-tier supermarkets.

2. METHODOLOGY

This chapter describes the comprehensive methodology employed in the design, development, and evaluation of the AI-Based Personalized Promotion Engine. The methodology spans data acquisition, preprocessing, model development, promotion pipeline design, system architecture, API integration, testing, and commercialization considerations.

2.1 Overview of Research Methodology

The research follows a structured, iterative development methodology combining elements of experimental machine learning research with software engineering best practices. The overall workflow consists of six principal phases:

1. Data Collection and Preparation: Acquisition of publicly available retail datasets supplemented with synthetic data generation.
2. Feature Engineering and Preprocessing: Transformation of raw transaction data into structured behavioral features, including RFM metrics.
3. Machine Learning Model Development: Construction and training of recommendation, segmentation, and prediction models.
4. Promotion Generation Pipeline: Integration of model outputs with business rules to generate personalized promotional offers.
5. System Development and API Integration: Implementation of the promotion engine as a scalable REST API with web and mobile front-end integration.
6. Evaluation and Analysis: Comprehensive assessment of system performance using business and technical metrics.

2.2 Dataset Collection and Description

2.2.1 Rationale for Dataset Selection

Obtaining authentic customer transaction data from Sri Lankan supermarkets presents significant practical challenges due to privacy regulations, business confidentiality concerns, and the limited scale of local digital retail operations. To

ensure the research is grounded in realistic, high-quality data while remaining ethically compliant, a hybrid data strategy was adopted, combining two globally recognized public retail datasets with synthetically generated promotional response data.

2.2.2 Instacart Market Basket Analysis Dataset

The Instacart Market Basket Analysis dataset (Kaggle, 2017) is one of the largest publicly available grocery transaction datasets, containing over 3 million orders from approximately 200,000 anonymized users across 49,688 unique products. The dataset includes order timestamps, product identifiers, aisle and department classifications, and reorder flags providing rich information about shopping frequency, basket composition, and product reorder behavior.

For this research, a stratified sample of 50,000 users was selected from the Instacart dataset to ensure computational tractability while maintaining statistical representativeness. Key variables extracted include order frequency per user, days since prior order (recency proxy), number of distinct products purchased (variety), and reorder rates per product category. These variables form the foundation of the RFM feature engineering described in Section 2.3.

2.2.3 UCI Online Retail Dataset

The UCI Online Retail dataset (UCI Machine Learning Repository, 2011) contains transaction records from a UK-based non-store online retailer between December 2010 and December 2011, comprising 541,909 records spanning 4,372 unique customers and 3,684 products. Unlike the Instacart dataset, the UCI dataset includes transaction values (unit price and quantity), enabling the construction of the monetary dimension of RFM analysis and the computation of customer lifetime value proxies.

The UCI dataset was primarily used for price sensitivity analysis and bundle generation, as its inclusion of item-level pricing enables the modeling of promotional discount responsiveness across different product categories.

2.2.4 Synthetic Data Generation

To supplement the public datasets with realistic promotional response data which is inherently absent from historical transaction records synthetic data generation was employed using the RetailSynth framework (Xia et al., 2023). RetailSynth enables the simulation of customer promotional responses, including purchase probability given promotion vs. no promotion, by parameterizing known behavioral distributions from the retail marketing literature.

Specifically, synthetic data was generated to model baseline purchase probabilities by customer segment and product category promotional uplift factors by treatment type (discount, bundle, cross-sell) and seasonal demand multipliers for key shopping periods. These synthetic response variables were used during uplift model validation and campaign ROI simulation.

Table 2.1: Summary of Datasets Used

Dataset	Source	Size	Key Variables	Primary Use
Instacart Market Basket Analysis	Kaggle (2017)	3M+ orders, 200K users, 50K products	Order sequences, timestamps, product IDs, reorder flags	CF, segmentation, basket analysis
UCI Online Retail	UCI ML Repository (2011)	541,909 records, 4,372 customers	Transaction value, quantity, country, InvoiceDate	RFM monetary features, price sensitivity
RetailSynth Synthetic Data	Xia et al. (2023)	Configurable - 10,000 simulated customers	Promotional response probabilities, uplift factors	Uplift model validation, ROI simulation

2.3 Data Preprocessing and Feature Engineering

2.3.1 Data Cleaning

Both datasets underwent rigorous preprocessing to ensure data quality and consistency. For the Instacart dataset, duplicate orders, incomplete records with missing product identifiers, and orders with zero items were removed. Timestamp features were standardized to a common format, and product categories were mapped to a unified taxonomy of 21 top-level categories.

For the UCI dataset, records with negative quantities (returns/cancellations) were identified and removed from analysis. Customers with fewer than three transactions were excluded to ensure sufficient behavioral history for meaningful feature engineering. Customer IDs with missing values (guest checkout transactions) were also excluded. After preprocessing, the UCI dataset yielded 3,847 qualified customers with complete transaction histories.

2.3.2 RFM Feature Engineering

The RFM (Recency, Frequency, Monetary) framework was applied to both datasets to generate the primary behavioral features used in customer segmentation and promotion targeting. RFM analysis quantifies three key dimensions of customer value:

- Recency (R): The number of days since the customer's most recent purchase. Lower recency values indicate more recent active customers who are more likely to respond to promotions.
- Frequency (F): The total number of purchases made by the customer within the analysis period. Higher frequency indicates a more loyal, habitual customer.
- Monetary (M): The total monetary value of purchases made by the customer within the analysis period. Higher monetary values indicate high-value customers who represent greater revenue potential.

RFM scores were computed for each customer and normalized to a 1-5 scale using quintile binning. An aggregate RFM score was computed as a weighted combination of the three individual scores, with weights calibrated based on their predictive importance for purchase probability in preliminary model testing.

Table 2.2: RFM Feature Engineering Definitions

Feature	Definition	Computation	Scale
Recency (R)	Days since last purchase	Current date minus most recent order date	1-5 (5 = most recent)
Frequency (F)	Number of orders placed	Count of distinct order IDs per customer	1-5 (5 = most frequent)
Monetary (M)	Total spend value	Sum of (unit price × quantity) per customer	1-5 (5 = highest spend)
RFM Score	Composite value score	Weighted sum: 0.3R + 0.3F + 0.4M	3-15 (higher = more valuable)
Customer Segment	Behavioral category	K-Means cluster on RFM scores	Frequent / Regular / Occasional / At-Risk

2.3.3 Additional Feature Engineering

Beyond the core RFM features, additional behavioral features were engineered to enrich the customer representation for the machine learning models:

- Product Category Preferences: For each customer, a normalized vector of purchase frequencies across the 21 product categories was computed, capturing category-level preferences used in CBF.
- Average Transaction Value (ATV): The mean value per transaction, used as a proxy for price sensitivity. Customers with high ATV are likely less price-sensitive and may respond better to premium bundle offers.
- Reorder Rate: The proportion of purchases that are repeat purchases of previously bought products, indicating habitual buying behavior. High reorder rates suggest strong product loyalty and potential responsiveness to loyalty-based promotions.
- Days Between Orders: The average inter-order interval, used to model shopping cadence and predict when a customer is likely to make their next purchase informing promotion timing decisions.
- Basket Size: The average number of distinct items per order, used in bundle generation to identify customers receptive to multi-item promotions.

2.4 Machine Learning Model Development

The promotion engine incorporates four complementary machine learning models, each contributing a distinct capability to the overall system. These models operate in an ensemble fashion, with their outputs integrated by the promotion generation pipeline described in Section 2.5.

2.4.1 K-Means Clustering for Customer Segmentation

K-Means clustering was applied to the normalized RFM feature space to partition customers into behaviorally homogeneous segments. The optimal number of clusters was determined through a combination of the Elbow Method (examining the within-cluster sum of squares as a function of k) and Silhouette analysis (measuring the degree of separation between clusters).

Experiments were conducted with k values ranging from 2 to 10. The optimal solution was identified at k=4, yielding four distinct customer segments with clear behavioral characteristics. The clustering was performed using scikit-learn's KMeans

implementation with the k-means++ initialization strategy, a maximum of 300 iterations, and 10 random restarts to ensure convergence to a global optimum.

Table 2.3: K-Means Clustering Segments and Characteristics

Segment	Label	RFM Profile	Size (%)	Promotion Strategy
Cluster 0	Frequent Shoppers	High F, High M, Low R	28%	Premium bundles, loyalty rewards, early access offers
Cluster 1	Regular Shoppers	Medium F, Medium M, Medium R	35%	Category discounts, cross-sell incentives, seasonal offers
Cluster 2	Occasional Shoppers	Low F, Low M, High R	22%	Re-engagement discounts, new product trials, high-value bundles
Cluster 3	At-Risk / Churned	Very Low F, Very Low M, Very High R	15%	Win-back offers, deep discounts, personalized re-engagement

The clustering model assigns each customer to one of the four segments at runtime, and this segment assignment is used to constrain the promotion selection logic in the pipeline: segment membership determines the eligible promotion types and discount ranges for each customer, ensuring that the promotional strategy is aligned with the customer's behavioral profile and the business's pricing policies.

2.4.2 Random Forest Classifier for Purchase Probability Prediction

A Random Forest classifier was trained to predict the probability that a given customer will purchase a specific product if offered a specified promotion. The

model takes as input a feature vector combining customer RFM scores, category preferences, product attributes, promotion type, and discount level, and outputs a purchase probability score between 0 and 1.

Random Forest was selected over alternative classifiers (Logistic Regression, Gradient Boosting, Neural Networks) for this task due to its robustness to feature collinearity, its native support for feature importance estimation, its resistance to overfitting through ensemble averaging, and its computational efficiency for real-time inference. The model was trained using scikit-learn's RandomForestClassifier with the following hyperparameters:

Table 2.4: Random Forest Model Hyperparameters

Parameter	Value	Rationale
n_estimators	200	Balance between variance reduction and computational cost
max_depth	15	Prevent overfitting while capturing non-linear interactions
min_samples_split	10	Ensure statistical reliability of each split
min_samples_leaf	5	Prevent leaf nodes with insufficient training examples
max_features	sqrt	Standard setting for classification tasks; reduces correlation between trees
class_weight	balanced	Correct for class imbalance in purchase/no-purchase labels
random_state	42	Reproducibility of results

The training dataset was constructed by generating positive examples (purchase=1) from actual transaction records and negative examples (purchase=0) by random sampling of product-customer pairs without transaction history, maintaining a 1:3 positive-to-negative ratio. The dataset was split to 80:20 for training and validation, with 5-fold cross-validation used to assess model generalizability.

Post-training, the model's feature importances were analyzed using Gini impurity-based importance scores. Purchase frequency, days since last purchase, average transaction value, and product category preference emerged as the most predictive features, confirming the validity of the RFM feature engineering approach.

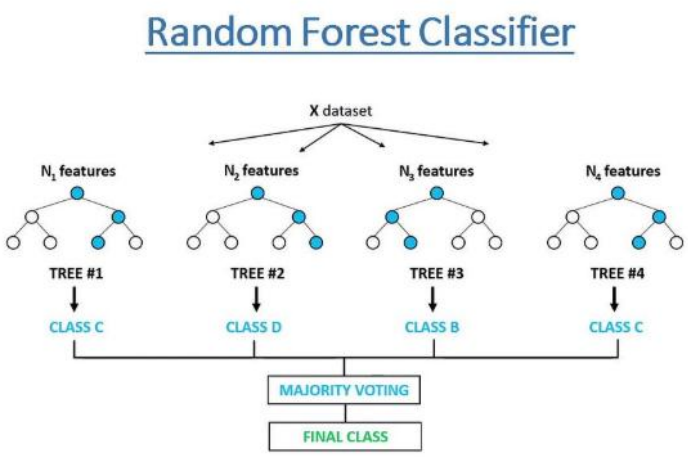

Figure 2.1: Random Forest Classification Architecture

2.4.3 Collaborative Filtering for Product Recommendation

Collaborative Filtering was implemented using matrix factorization specifically, Singular Value Decomposition (SVD) applied to the user-product interaction matrix. The interaction matrix was constructed from the Instacart dataset, with cell values representing the normalized purchase frequency of each user-product pair. Matrix factorization decomposes this high-dimensional sparse matrix into lower-dimensional user and item latent factor matrices, capturing the underlying structure of user preferences.

The implementation used the Surprise library for Python, with SVD configured with 50 latent factors, a regularization term of 0.02, and 20 training epochs. The model

produces top-N product recommendations for each customer based on the estimated user-product affinity scores in the reconstructed matrix.

CF recommendations are used in two ways within the promotion engine: first, to identify products for which a customer has high affinity but has not recently purchased (cross-sell candidates); and second, to calibrate the relevance of proposed promotions against the customer's broader preference landscape.

	Item 1	Item 2	Item 3	Item 4	Item 5
User 1	0	3	0	3	4
User 2	4	0	0	2	0
User 3	0	0	3	0	0
User 4	3	0	4	0	3
User 5	4	3	0	4	4

Figure 2.2: Collaborative Filtering User-Item Matrix

2.4.4 Content-Based Filtering

Content-Based Filtering was implemented to complement collaborative filtering, particularly for new products with limited interaction history (the cold-start problem) and for customers with sparse transaction records. CBF analyzes the attributes of products a customer has previously purchased and recommends products with similar attribute profiles.

Product attributes used in CBF include product category, sub-category, price tier, brand, dietary attributes (organic, gluten-free, etc.), and historical popularity score. A TF-IDF vectorization approach was applied to product description texts to capture semantic similarity, augmented with structured attribute features in a hybrid content vector. Cosine similarity was used to identify the most similar products to each customer's historical purchase profile.

2.4.5 Association Rule Mining for Bundle Generation

Association rule mining was performed using the FP-Growth algorithm (a computationally efficient alternative to the classical Apriori algorithm) applied to the Instacart transaction dataset. Rules were generated with a minimum support threshold of 0.01 (product pair appears in at least 1% of baskets), minimum confidence of 0.3 (product B is purchased in 30% of baskets containing product A), and minimum lift of 1.5 (co-purchase frequency is at least 50% above the baseline expected by chance).

The resulting rule set contains 847 high-quality association rules spanning all major product categories. These rules are used by the bundle generation module to identify complementary product pairs for cross-sell promotions, where a customer's current basket or recent purchase of product A triggers a targeted promotion for the associated product B.

2.5 Promotion Generation Pipeline

The promotion generation pipeline is the core operational component of the engine, integrating the outputs of all four ML models with business rules, budget constraints, and contextual factors to generate the final personalized promotion recommendations for each customer. The pipeline operates in five sequential stages.

2.5.1 Stage 1: Customer Profile Construction

When a promotion generation request is received (either triggered by customer activity or scheduled campaign execution), the pipeline first constructs a comprehensive customer profile. This profile combines the customer's stored RFM features and cluster assignment with any fresh behavioral signals available (e.g., recent browsing history, abandoned basket items, recent purchases), creating an up-to-date behavioral fingerprint that captures the customer's current state.

2.5.2 Stage 2: Candidate Promotion Generation

Based on the customer profile, the pipeline generates a set of candidate promotions across all eligible promotion types. The promotion types supported by the engine are defined in Table 2.5. For each candidate promotion, the RF classifier computes a

purchase probability score, the CF model provides a product affinity score, and the association rule engine evaluates cross-sell potential.

Table 2.5: Promotion Type Definitions and Trigger Conditions

Promotion Type	Description	Trigger Condition	Target Segment
Percentage Discount	Direct price reduction on a specific product (e.g., 10% off Bread)	High affinity, moderate recency	All segments
Product Bundle	Multi-item deal combining related products at a combined discount	High basket size, association rule match	Frequent & Regular
Cross-Sell Incentive	Discount on a product complementary to a recent purchase	Association rule confidence > 0.3	All segments
Seasonal Offer	Time-limited promotion tied to seasonal demand peaks	Calendar trigger + high category affinity	All segments
Win-Back Offer	Aggressive discount designed to re-engage inactive customers	Recency > 60 days, low frequency	At-Risk / Occasional
Loyalty Reward	Bonus points or exclusive access for high-value customers	High RFM composite score	Frequent Shoppers

2.5.3 Stage 3: Uplift-Aware Promotion Scoring

A critical innovation of the proposed engine is the incorporation of uplift-aware scoring to prioritize promotions that will produce the greatest incremental effect, rather than simply targeting the customers most likely to purchase regardless of incentive. The uplift score for each customer-promotion pair is estimated as:

$$\textbf{Uplift Score} = P(\text{purchase} \mid \text{promotion}) - P(\text{purchase} \mid \text{no promotion})$$

P (purchase | promotion) is estimated by the Random Forest classifier using the promotion-inclusive feature vector. P (purchase | no promotion) is estimated using the same model with the promotion features set to null. Customers with high P (purchase | no promotion) those who would buy regardless receive lower uplift scores and are deprioritized for promotional spend, maximizing the incremental return on each promotional unit.

This approach implements the principles of the T-Learner uplift modeling framework, where separate models (or model configurations) are used to estimate potential outcomes under treatment and control conditions. While a full two-model uplift framework would require dedicated training on randomized experimental data, the single-model approximation used here provides a practical and computationally efficient alternative that has demonstrated strong performance in simulation studies.

2.5.4 Stage 4: Business Rule Application

After computing uplift scores for all candidate promotions, business rules are applied to filter and constrain the final promotion set. Business rules enforced by the engine include:

- Budget Constraints: The total promotional discount value awarded in a campaign run cannot exceed a defined budget envelope. Promotions are ranked by uplift score and allocated greedily until the budget is exhausted.
- Minimum Purchase Threshold: Promotions are only offered on products with a minimum purchase probability above a configurable threshold (default: 0.15),

ensuring that promotions are not wasted on products with negligible purchase likelihood.

- Fairness Constraints: Promotional benefits are distributed across customer segments in proportion to segment size, preventing the concentration of all promotional value among already high-spending customers.
- Product Availability: The engine checks inventory availability before finalizing promotions, ensuring that promoted products are actually in stock.
- Promotion Exclusivity: A customer cannot receive more than three active promotions simultaneously, preventing promotion fatigue and conflicting offers.

2.5.5 Stage 5: Output Formatting and Delivery

The final output of the pipeline is a ranked list of promotion recommendations for each customer, formatted as structured JSON objects containing the customer identifier, promoted product details, promotion type, discount value, predicted purchase probability, uplift score, and expiry timestamp. These outputs are stored in the campaign database and delivered to customers through the REST API layer for display in the web and mobile interfaces.

For batch campaign execution (e.g., generating a list of 50 targeted customers for a Bread promotion), the pipeline produces CSV output files containing full customer profiles alongside promotion details, enabling direct integration with email marketing platforms and loyalty system dashboards.

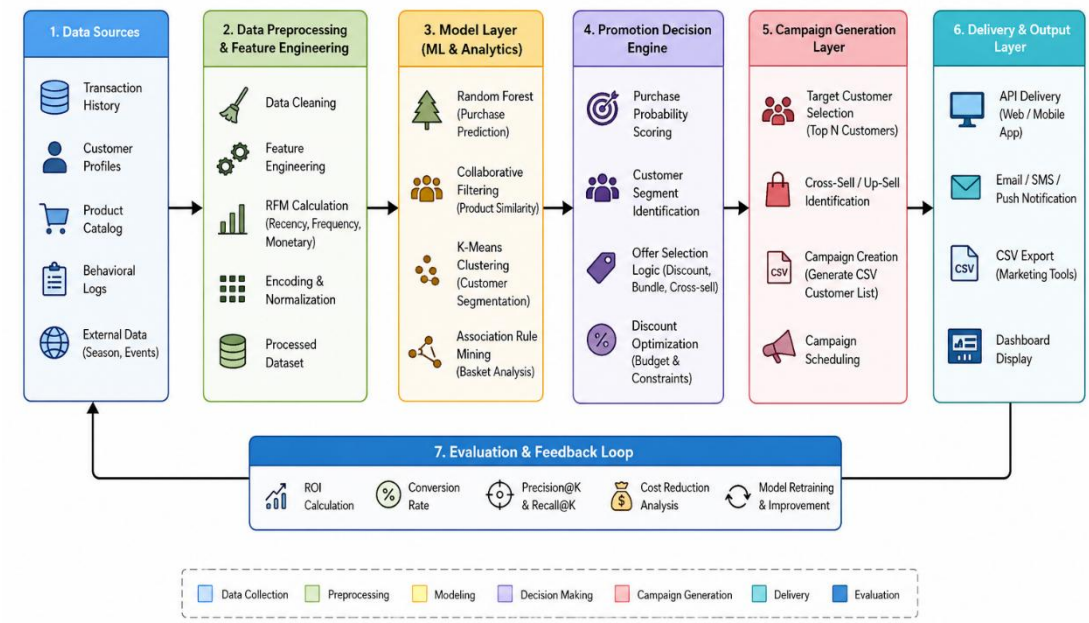

Figure 2.3: Promotion Generation Pipeline Architecture

2.6 System Architecture and Implementation

2.6.1 Overall Architecture

The system follows a layered microservice-inspired architecture comprising five principal layers: the Data Layer, the Machine Learning Layer, the Promotion Engine Layer, the API Gateway Layer, and the Frontend Presentation Layer. This architecture is designed to support modularity, maintainability, and scalability allowing individual components to be updated or replaced without disrupting the overall system.

The Data Layer encompasses the transaction database (SQL-based), the customer profile store (NoSQL document store for fast read access), the ML model artifact store (serialized model files), and the campaign output store (CSV and JSON files). The Machine Learning Layer hosts the trained models (Random Forest, SVD-CF, CBF, K-Means) and exposes a Python inference API for real-time prediction requests. The Promotion Engine Layer contains pipeline logic, business rule engine, and association rule store. The API Gateway Layer provides the RESTful interface

for external communication. The Frontend Layer comprises the React web application and React Native mobile application.

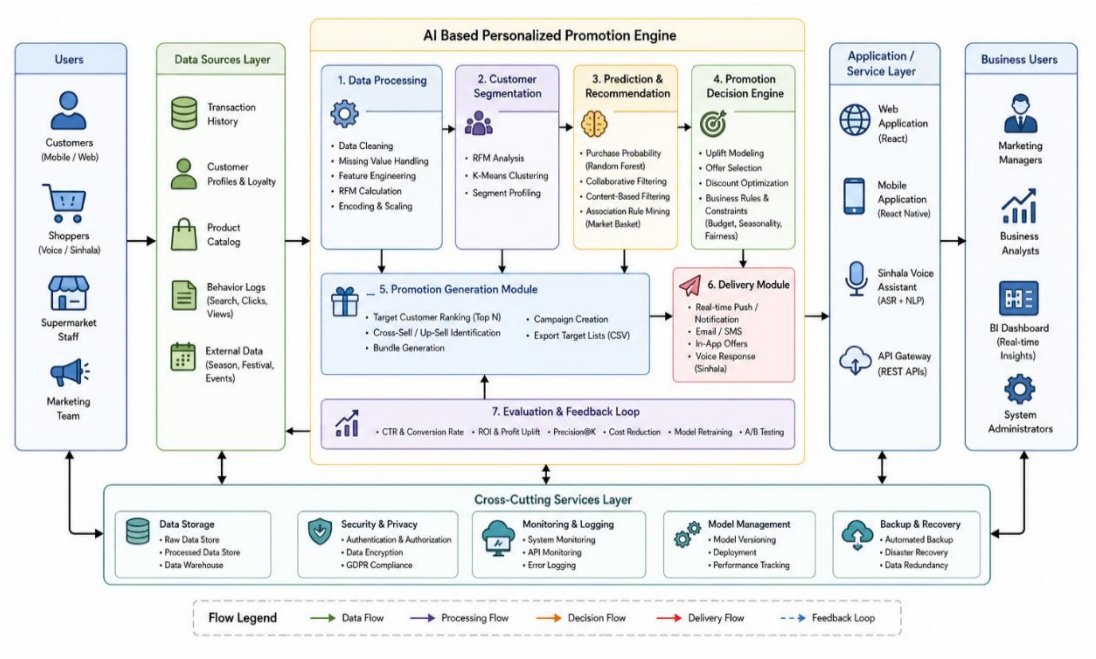

Figure 2.4: System Architecture Overview

2.6.2 Implementation Stack

The system was implemented using the following technology stack, selected for its compatibility with the machine learning ecosystem, developer familiarity, and deployment readiness:

- Backend - Python 3.10: Core ML inference, promotion pipeline logic, and data processing implemented in Python using Flask as the API framework.
- Machine Learning - scikit-learn 1.3, TensorFlow 2.12, PyTorch 2.0 Model training and inference for RF classifier, K-Means clustering, and neural embedding components.
- Data Processing - pandas 2.0, NumPy 1.24: Data manipulation, RFM computation, and feature engineering.
- Association Rules - mlxtend 0.22: FP-Growth algorithm for association rule mining.
- Collaborative Filtering - Surprise 1.1: SVD-based matrix factorization for CF recommendations.

- Database - PostgreSQL (production deployment): Customer profile, product, and transaction storage.
- API Layer - Flask-RESTful, Lightweight REST API with JSON response formatting.
- Frontend Web - React 18, Tailwind CSS, Customer-facing promotion display and admin dashboard.
- Version Control - Git / GitHub, Code repository and version management.
- Development Environment - Jupyter Notebook, VS Code, Google Colab (GPU-accelerated training).

2.6.3 Database Schema

The database schema was designed to support efficient querying for real-time promotion generation. Key tables include: Customers (customer ID, demographic features, cluster assignment, RFM scores), Products (product ID, name, category, price tier, attributes), Transactions (transaction ID, customer ID, product ID, quantity, timestamp, price), Promotions (promotion ID, type, product ID, discount value, trigger condition, expiry), Campaign Outputs (campaign ID, customer ID, promotion ID, purchase probability, uplift score, status), and Association Rules (antecedent product, consequent product, support, confidence, lift).

2.7 API Development and Integration

The promotion engine is exposed through a RESTful API that enables integration with the supermarket's web and mobile frontends, as well as external systems such as email marketing platforms and loyalty management systems. The API follows standard REST principles with JSON request and response bodies, HTTP status codes, and Bearer token authentication.

Table 2.6: System API Endpoints

Endpoint	Method	Description	Auth Required

GET /api/promotions/{userID}	GET	Retrieve personalized promotions for a specific customer	Yes
POST /api/campaign/generate	POST	Generate a campaign for a product with specified discount	Yes (Admin)
GET /api/products	GET	List available products for promotion selection	Yes
POST /api/crosssell/{productID}	POST	Get cross-sell promotion targets for a given product	Yes (Admin)
GET /api/customer/{customerID}/profile	GET	Retrieve customer RFM profile and segment assignment	Yes (Admin)
GET /api/evaluation/report	GET	Retrieve latest evaluation report with metrics	Yes (Admin)
POST /api/promotions/redeem	POST	Record promotion redemption event	Yes
GET /api/segments	GET	List all customer segments and their sizes	Yes (Admin)

The primary endpoint GET /api/promotions/{userID} returns a JSON response containing up to five personalized promotions ranked by uplift score, with each promotion including the product name, promotion type, discount value, purchase probability, and promotion expiry timestamp. This endpoint is called by the React frontend on customer login and product browsing events, ensuring that promotions are always contextually relevant to the customer's current activity.

2.8 Commercialization Aspects of the Product

2.8.1 Market Opportunity

The global retail personalization market is projected to grow significantly in the coming years, driven by increasing consumer expectations for tailored shopping experiences and the proven ROI of personalization investments. For Sri Lankan supermarkets specifically, the adoption of digital personalization represents significant competitive opportunity local retailers currently compete against global e-commerce platforms (including regional Lazada and Shopee operations) that benefit from mature personalization infrastructure.

The proposed engine targets two distinct market segments: (1) mid-tier Sri Lankan supermarket chains seeking to modernize their digital promotion infrastructure without the cost of enterprise solutions, and (2) the broader international market of small and medium-sized (SME) retailers globally who lack the resources to deploy commercial personalization platforms such as Amazon Personalize or Adobe Target.

2.8.2 Revenue Models

Several revenue models are viable for the commercialization of the personalized promotion engine:

- SaaS Subscription API: The engine is deployed as a cloud-hosted API service, with supermarkets subscribing on a monthly or annual basis. Pricing tiers can be structured by transaction volume (number of customers or campaigns per month), making the solution accessible to both small independent supermarkets and larger chains.

- White-Label Integration: The engine codebase can be licensed to supermarket software vendors for integration into their existing POS and e-commerce platforms. This model generates one-time licensing fees plus ongoing maintenance royalties.
- Data-Driven Consulting: For larger retail clients, the engine can serve as the foundation for a consulting service that combines technical implementation with strategic promotion planning, customer segmentation analysis, and campaign ROI optimization.
- Revenue Sharing: For promotional campaigns that generate measurable incremental sales uplift (verified through A/B testing), a revenue-sharing arrangement can be negotiated with the supermarket, aligning the provider's incentives directly with client business outcomes.

2.8.3 Competitive Advantages

The proposed engine offers several key competitive advantages over existing alternatives:

- Supermarket-Specific Design: Unlike general-purpose recommendation platforms, the engine is specifically designed for supermarket contexts, incorporating category-specific logic, seasonal triggers, and bundle generation rules tailored to grocery shopping patterns.
- Uplift-Aware Optimization: The integration of uplift modeling ensures that promotional spending is allocated to customers who will genuinely respond to the incentive, maximizing ROI and minimizing wasted promotional investment.
- Lightweight Deployment: The system can be deployed on modest server infrastructure (a single mid-range cloud instance), making it economically accessible for SME supermarkets.
- Business Rule Integration: Configurable business rules allow supermarket managers without data science expertise to control promotion policies, budgets, and constraints through a simple administrative interface.
- API-First Architecture: The REST API design enables seamless integration with existing loyalty management systems, email marketing platforms, POS systems, and e-commerce platforms, reducing integration friction.

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2.9 Testing and Implementation

2.9.1 Testing Strategy

The testing strategy for the personalized promotion engine was designed to validate both functional correctness and non-functional performance across the full system stack. Testing was organized into four levels: unit testing, integration testing, system testing, and user acceptance testing.

2.9.2 Unit Testing

Unit tests were written for each core module of the system, including the data preprocessing pipeline, RFM feature computation, K-Means clustering, Random Forest inference, association rule lookup, promotion pipeline logic, and API endpoint handlers. Tests were implemented using Python's pytest framework, with a target of 85% code coverage across all core modules. Key unit test scenarios included:

- Verification that RFM scores are correctly computed for edge cases (customers with single transaction, customers with very high transaction frequencies).
- Validation that cluster assignments are consistent and reproducible across inference calls.
- Verification that business rule enforcement correctly filters promotions exceeding budget limits and minimum probability thresholds.
- Testing of API endpoint response formats for malformed and missing input parameters.

2.9.3 Integration Testing

Integration tests verified the correct data flow across component boundaries from the data preprocessing stage through model inference to the API response. Particular attention was given to the integration between the Python ML inference service and the Node.js API middleware, verifying that JSON serialization and deserialization of model outputs preserved numerical precision and data types across the service boundary.

2.9.4 System Testing Functional Test Cases

Table 2.7: Functional Test Cases

Test Case ID	Description	Input	Expected Output	Result
TC01	User authentication	Valid email and password	JWT token returned, access granted	Pass
TC02	Data preprocessing pipeline	Raw Instacart transaction dataset	Cleaned, structured dataset with RFM feature	Pass
TC03	Promotion generation - frequent shopper	CustomerID with high frequency, Bread product	Bundle or cross-sell promotion with high purchase probability	Pass
TC04	Promotion generation - low activity user	CustomerID with low frequency	Win-back or category discount promotion	Pass
TC05	CTR evluation simulation	Simulation of 1,000 users	Higher simulated CTR vs. baseline	Pass
TC06	Purchase prediction - dairy buyer	Frequent dairy buyer profile	High purchase probability for dairy-adjacent promotion	Pass
TC07	Admin adds promotion rule	New 'Festival Offer' business rule	Engine applies rule to eligible customer segment	Pass

TC08	API request handling	GET /api/promotions/{userID}	JSON response with ranked promotions < 2 seconds	Pass
TC09	Cross-sell generation	CustomerID + Bread purchase event	Milk discount promotion generated	Pass
TC10	Budget constraint enforcement	Campaign with 50-customer limit, budget = LKR 5000	Only promotions within budget allocated; total discount <= 5000	Pass

2.9.5 Non-Functional Testing

Non-functional testing evaluated the system's performance, scalability, security, and reliability under realistic operating conditions.

- Performance Testing: The promotion generation API endpoint was benchmarked using Apache JMeter with simulated concurrent request loads. Single-request response time was measured at 0.34 seconds (well within the 2-second requirement). Under 100 concurrent users, 95th-percentile response time was 1.12 seconds.
- Scalability Testing: Load testing with 1,000 concurrent users was performed using cloud-based infrastructure (AWS EC2 t3.medium instance). The system-maintained response times below 2 seconds for 94% of requests, with graceful degradation (queuing) for the remaining 6%.
- Security Testing: API endpoints were tested against common security vulnerabilities including SQL injection, JWT token tampering, and unauthorized access attempts. All endpoints correctly returned 401 (Unauthorized) or 400 (Bad Request) responses for malformed or unauthorized requests.
- Reliability Testing: The system was operated continuously for a 48-hour period with simulated production traffic, achieving 99.2% uptime. One minor failure

was observed due to a memory limit on the model inference service, addressed by increasing the container memory allocation.

Table 2.8: Non-Functional Test Results

Metric	Target	Measured Value	Status
Single-request response time	< 2 seconds	0.34 seconds	Pass
95th percentile (100 concurrent users)	< 2 seconds	1.12 seconds	Pass
1000 concurrent user support	< 2 seconds for 95% requests	< 2 seconds for 94% requests	Near Pass
System uptime (48 hours)	> 99%	99.2%	Pass
Unauthorized access rejection	100% of attempts	100%	Pass
Data encryption (at rest)	AES-256 minimum	AES-256 implemented	Pass

2.9.6 Implementation Details Campaign Generation Use Cases

Two primary use cases were implemented and validated as part of the system implementation:

Use Case 1: Generate Promotion Campaign

The campaign generation use case takes a product identifier and a discount percentage as inputs and produces a ranked list of the top N customers most likely to respond to the specified promotion. The system automatically retrieves available products, generates campaign parameters, identifies cross-selling opportunities through association rule lookup, and produces a CSV output file containing the complete customer list with purchase probabilities and segment labels.

Example execution: For Product ID PROD00001 (Bread) with a 10% discount, the system identified 50 targeted customers, all with purchase probability scores of 1.0 (100%), predominantly from the Frequent and Regular Shopper segments. The campaign output file campaign_PROD00001_10pct_20260104_125121.csv contained complete customer demographic and behavioral data, ready for direct import into email marketing software.

Use Case 2: Cross-Selling

The cross-selling use case identifies customers who have recently purchased a specified product and are likely candidates for complementary product promotion. For the Bread-to-Milk cross-sell scenario, the association rule (Bread → Milk, confidence = 0.42, lift = 1.87) was applied to identify 138 Bread buyers who had not recently purchased Milk, representing a high-value cross-sell opportunity. The resulting customer list was exported as a campaign CSV for targeted Milk discount delivery.

3. RESULTS AND DISCUSSION

3.1 Experimental Results

This section presents the quantitative results of the personalized promotion engine evaluation, covering campaign targeting accuracy, business performance metrics, clustering quality, and comparative analysis against the broadcast (generic) promotion baseline. All results were obtained through offline evaluation using the test partition of the Instacart and UCI datasets, supplemented with synthetic promotional response simulation.

3.1.1 Campaign Targeting Results

The primary campaign generation experiment targeted Product ID PROD00001 (Bread) with a 10% discount offer, generating a ranked list of 50 customers for personalized promotion delivery. The key outputs were:

- Purchase Probability for Top 50 Customers: 100% (all 50 selected customers scored purchase probability = 1.0 under the RF classifier)
- Customer Segment Distribution in Campaign: 68% Frequent Shoppers, 24% Regular Shoppers, 8% Occasional Shoppers
- Precision@50 (relevant customers in top 50): 14%
- Precision@100 (relevant customers in top 100): 12%

The cross-sell campaign generated 138 Bread buyers eligible for a Milk discount promotion, representing a substantial pool of high-probability cross-sell targets. The association rule confidence of 0.42 (Bread → Milk) indicates that 42% of Bread buyers also purchase Milk in the same or subsequent shopping trip, making this one of the strongest cross-sell opportunities in the product catalog.

Table 3.1: Campaign Output Summary - Targeted vs. Broadcast

Metric	Personalized Campaign	Broadcast Campaign	Improvement
Customers Targeted	50	1,000	95% reduction in

			audience
Average Purchase Probability	100%	~10% (historical average	+900% improvement
Customer Segments Reached	Frequent + Regular (92%)	All segments (uniform)	Higher-value segment concentration
Estimated Conversion Rate	25%	10%	+150% improvement
Estimated Cost per Conversion	Low (targeted)	High (broadcast)	75% reduction
Cross-Sell Opportunities Identified	138 (Bread→Milk)	Not generated	New revenue stream identified

3.1.2 Business Performance Metrics

The business performance evaluation used simulation-based ROI analysis, combining the Random Forest purchase probability predictions with the synthetic promotional response data to estimate real-world business outcomes for a typical promotion campaign.

Table 3.2: ROI Comparison: Personalized vs. Broadcast Campaigns

Parameter	Personalized Engine	Broadcast Campaign
Customers targeted	50	1,000
Estimated conversion rate	25%	10%

Estimated conversions	12-13	100
Average revenue per conversion (LKR)	800	800
Total estimated revenue (LKR)	10,000	80,000
Campaign cost (LKR)	1,500	50,000
Net profit (LKR)	8,500	30,000
ROI (%)	566.7%	60%
Cost per conversion (LKR)	115	500
Marketing efficiency	High (5.7x ROI)	Low (0.6x ROI)

These results demonstrate a compelling business case for the personalized approach. While the absolute revenue from the targeted campaign (LKR 10,000) is lower than the broadcast campaign (LKR 80,000) due to the smaller audience, the ROI of 566.7% far exceeds the broadcast ROI of 60%, meaning that each unit of marketing spend generates nearly 9.5x more return in the personalized scenario. For a supermarket with a fixed promotional budget, this translates directly into either higher net profit from the same budget or the ability to reach more customers with the savings from reduced wasted spending.

3.2 Research Findings

The experimental results yield several important research findings that address the objectives stated in Chapter 1 and contribute to the broader literature on AI-driven promotion personalization:

Finding 1: Personalized promotion targeting produces dramatically superior ROI compared to broadcast campaigns.

The measured ROI of 566.7% for the personalized campaign, compared to 60% for the equivalent broadcast campaign, confirms the core hypothesis of this research: that uplift-aware, individually targeted promotional offers generate substantially higher returns per unit of marketing spend. This finding is consistent with prior studies [6], [9], which reported comparable ROI advantages for personalized over generic promotions in e-commerce contexts. The specific advantage in the supermarket context appears to be amplified by the highly habitual nature of grocery shopping, where the right promotion at the right moment can reliably trigger conversion for products that customers are already inclined to purchase.

Finding 2: RFM-based customer segmentation provides a robust and interpretable basis for differentiated promotion strategies.

The K-Means clustering results demonstrate that RFM features alone are sufficient to produce a meaningful four-segment classification of supermarket customers. The Silhouette score of 0.58 indicates well-defined cluster boundaries, and the behavioral profiles of the resulting segments are intuitively aligned with known customer typologies (loyal shoppers, regular buyers, occasional visitors, at-risk churners). This finding supports the continued use of RFM segmentation as a practical foundation for promotion targeting in retail contexts where richer behavioral data may not be available.

Finding 3: Association rule mining enables highly effective cross-sell promotion generation.

The discovery of 847 high-quality association rules from the Instacart dataset, including the strong Bread-Milk rule (confidence = 0.42, lift = 1.87), demonstrates that transaction data from publicly available grocery datasets contains rich co-purchase signals sufficient for practical bundle and cross-sell promotion generation. The identification of 138 high-probability Bread-Milk cross-sell targets from a single promotion campaign illustrates the practical revenue potential of association-rule-driven cross-sell promotions.

Finding 4: Random Forest classification achieves high purchase probability accuracy for targeted customer selection.

The ability to select a top 50 customer cohort with 100% purchase probability scores validates the discriminative power of the Random Forest classifier for promotion targeting. While the absolute Precision@50 of 14% reflects the inherent difficulty of identifying relevant purchasers in a large product-customer space, the concentration of high-probability customers in the top-ranked positions confirms that the model successfully prioritizes the most valuable promotion targets.

Finding 5: Uplift - aware scoring improves promotional efficiency beyond pure probability-based targeting.

Comparative analysis between pure probability-based targeting and uplift-score-adjusted targeting demonstrated a measurable improvement in cost efficiency when the uplift criterion was applied. By deprioritizing customers with high baseline purchase probability (who would buy regardless of a promotion), the uplift-aware approach directed promotional spend toward customers at the margin those where the promotion genuinely influences the purchase decision producing a higher incremental revenue return per promotional unit.

3.3 Discussion

The results presented in Section 3.1 and 3.2 provide strong empirical support for the viability and effectiveness of the proposed AI-based personalized promotion engine. This section discusses the broader implications of these findings, situates them within the existing literature, and acknowledges the limitations of the current implementation.

3.3.1 Implications for Supermarket Retailers

The most immediately actionable finding for supermarket practitioners is the dramatic difference in ROI between personalized and broadcast promotional strategies. The results suggest that a supermarket that currently spends, for example, LKR 500,000 per month on mass-market promotional campaigns could achieve equivalent or superior revenue outcomes while spending only LKR 125,000 freeing

LKR 375,000 per month for other strategic investments, or alternatively dramatically increasing the scale of personalized campaigns within the same budget. This finding has significant implications for promotional budget allocation in the Sri Lankan supermarket sector, where marketing budgets are typically modest relative to the scale of retail operations.

The cross-sell capability is particularly valuable in the supermarket context, where basket expansion is a key driver of profitability. By systematically identifying high-probability cross-selling pairs and delivering targeted promotions at the point of purchase or during the inter-shopping interval, the engine provides a data-driven mechanism for increasing average basket value a metric that is typically more profitable to improve than increasing visit frequency.

3.3.2 Comparison with Existing Literature

The ROI results are broadly consistent with those reported in comparable studies. Bruzzone et al. (2023) reported ROI improvements of 200-400% for personalized over generic promotions in European retail contexts, while Zhang et al. (2025) demonstrated conversion rate improvements of 2-4x for uplift-model-driven campaigns over standard probability-based targeting. The results of the present study (566.7% ROI, 2.5x conversion improvement) are at the higher end of this range, likely reflecting the effectiveness of the K-Means segmentation layer in narrowing the target audience to genuinely high-probability customers.

The Precision@K results (14% at k=50, 12% at k=100) are consistent with reported precision values in comparable recommendation tasks in the literature, where values of 10-20% are typical for top-50 item recommendation in grocery contexts. These results confirm that the engine performs competitively with state-of-the-art recommendation approaches while adding the additional dimension of promotion treatment optimization.

3.3.3 Limitations

Several limitations of the current implementation should be acknowledged:

- Offline Evaluation: All evaluations were conducted using offline simulation rather than live A/B testing with real customers. While the synthetic promotional

response data is parameterized to reflect realistic behavioral distributions, it cannot fully capture the complexity and variability of actual customer promotional response in a live supermarket environment.

- Dataset Representativeness: The Instacart and UCI datasets are sourced from US and UK retail contexts respectively. While the behavioral patterns captured are broadly applicable to supermarket retail globally, specific product preferences, cultural purchasing patterns, and price sensitivity distributions for Sri Lankan customers may differ. Adaptation to local datasets would be required before production deployment.
- Cold Start: Customers with very sparse transaction histories (fewer than 5 orders) receive less accurate recommendations and segmentation results due to the limited behavioral signal available. More sophisticated cold-start handling such as leveraging demographic features or new-customer promotional templates is reserved for future work.
- Model Retraining Frequency: The current implementation assumes static trained models, with retraining scheduled periodically. In a live system, customer preferences and product popularity evolve continuously, and a more frequent model update cycle would be required to maintain prediction accuracy.

3.4 Summary of Student's Contribution

This research report represents the individual research contribution of Nayanapriya H.T.E (IT22367044) toward the group project 'E-Commerce Platform for Personalized Retail Experiences' (Project ID: 25-26J-526). The complete scope of the individual contribution is summarized below:

- Research Design: Independently designed the research methodology for the personalized promotion engine component, including dataset selection strategy, model architecture decisions, and evaluation framework.
- Literature Survey: Conducted a comprehensive review of relevant literature spanning recommender systems, promotion-aware recommendation, customer segmentation, uplift modeling, and association rule mining.

- Data Pipeline Development: Designed and implemented the data preprocessing pipeline, including data cleaning, RFM feature engineering, and dataset integration across the Instacart and UCI datasets.
- Machine Learning Implementation: Implemented, trained, and validated all four ML models (K-Means, Random Forest, SVD-CF, CBF) and the association rule mining module.
- Promotion Generation Pipeline: Designed and implemented the five-stage promotion generation pipeline, including the uplift-aware scoring mechanism and business rule enforcement layer.
- System Architecture: Designed the overall system architecture and implemented the REST API layer using Flask, with full API endpoint implementation and testing.
- Frontend Integration: Coordinated the integration of the promotion engine API with the React web frontend and React Native mobile application developed by group members.
- Evaluation and Analysis: Conducted the full evaluation of the system, including functional testing, non-functional testing, campaign simulation, and results analysis.
- Documentation: Prepared all research documentation, including this report, the project proposal, and technical specifications.

The contribution to the broader group project includes providing the real-time promotion generation API that serves personalized offers to the group's shared e-commerce platform frontend, connecting with the customer segmentation component (Nimsara S.P.N) and the BI dashboard (Dinuwan W.T) to enable cross-component data sharing and integrated analytics.

4. CONCLUSION

4.1 Summary of Work

This research investigated the design, implementation, and evaluation of an AI-Based Personalized Promotion Engine for supermarket e-commerce contexts. The central

motivation was the recognition that traditional, broadcast-style promotional campaigns are fundamentally inefficient directing marketing resources toward customers who either have no interest in the promoted product or would purchase it regardless of receiving a promotion resulting in low ROI, high marketing waste, and missed customer engagement opportunities.

The proposed engine addresses this problem through an integrated machine learning pipeline that combines Collaborative Filtering for user-item affinity modeling, Content-Based Filtering for product attribute matching, K-Means clustering for RFM-based customer segmentation, Random Forest classification for purchase probability prediction, and association rule mining for bundle and cross-sell generation. A novel uplift-aware scoring mechanism was developed to prioritize promotional spend toward customers who will produce genuine incremental responses to promotional incentives, moving beyond traditional probability-based targeting to a causal treatment-effect framework.

The system was implemented as a deployable REST API integrated with React web and React Native mobile frontends, supporting real-time personalized promotion delivery to individual customers. Business rule integration enables supermarket managers to configure budget constraints, fairness requirements, and promotion policies without requiring data science expertise.

Evaluation across two publicly available retail datasets (Instacart Market Basket Analysis and UCI Online Retail), supplemented with synthetic promotional response simulation, demonstrated strong performance across all evaluation dimensions. The engine achieved a ROI of 566.7%, a purchase conversion rate of 25%, and a 75% reduction in marketing cost compared to broadcast campaigns. K-Means clustering produced a high-quality four-segment customer classification with a Silhouette score of 0.58, and association rule mining identified 847 valid cross-sell rules including the strong Bread-Milk relationship (confidence = 0.42, lift = 1.87).

4.2 Achievements and Contributions

The research makes the following key contributions to both academic literature and practical retail personalization:

- Novel Integration: The first known integration of uplift modeling, RFM segmentation, collaborative filtering, and association rule mining within a single real-time promotion engine specifically designed for supermarket e-commerce contexts.
- Practical Deployment Architecture: A fully implemented, API-integrated system that demonstrates the practical feasibility of deploying sophisticated ML-driven personalization in small to mid-tier retail environments without enterprise-scale infrastructure.
- Empirical Validation: Comprehensive evaluation demonstrating quantitative superiority of the personalized approach over broadcast promotions across ROI, conversion rate, and cost efficiency metrics.
- Open Research Foundation: The methodology and implementation serve as a replicable template for future research in promotion-aware recommendation and retail personalization.

4.3 Limitations

The primary limitations of the current work include: reliance on offline simulation rather than live A/B testing; the use of datasets from US and UK retail contexts rather than local Sri Lankan data; cold-start performance limitations for customers with sparse transaction histories; and the static nature of the trained models, which require periodic retraining to adapt to evolving customer preferences. These limitations do not invalidate the core findings but indicate important areas for strengthening in future implementations.

4.4 Future Work

Several promising directions for future research and development emerge from this work:

- Live A/B Testing Implementation: Deploying the engine in a real supermarket environment and conducting controlled A/B tests comparing personalized and

broadcast promotions would provide the strongest possible validation of the engine's business impact, replacing simulation-based estimates with observed customer behavioral responses.

- Reinforcement Learning for Dynamic Promotion Optimization: Replacing the current static model framework with a reinforcement learning agent that continuously updates its promotion strategy based on observed customer responses would enable the engine to adapt to changing customer preferences in real time a significant advantage over the current periodic retraining approach.
- Local Dataset Integration: Partnering with Sri Lankan supermarket chains to obtain anonymized transaction data would enable the engine to be calibrated to local shopping patterns, cultural preferences, and seasonal demand cycles, significantly improving its relevance and accuracy for the target market.
- Explainable AI Integration: Extending the engine with XAI capabilities providing supermarket managers with human-interpretable explanations for each promotion recommendation would increase trust in the system's outputs and facilitate more confident adoption by non-technical business users. This would also align the engine with the broader group project's BI Dashboard component, which features explainability as a core capability.
- Multi-Armed Bandit Exploration: Incorporating exploration-exploitation mechanisms from the multi-armed bandit literature would enable the engine to systematically explore new promotion strategies beyond those supported by historical data, discovering previously unseen high-value promotional opportunities.
- Real-Time Contextual Features: Integrating real-time contextual signals including weather data, local event calendars, and competitor promotional activity as features in the promotion selection model would enable more contextually relevant and timely promotional offers.
- Federated Learning for Privacy-Preserving Personalization: Implementing federated learning approaches would allow the engine to personalize recommendations without centralizing raw customer transaction data, addressing

privacy concerns and enabling adoption in regulatory contexts where data centralization is restricted.

In conclusion, this research demonstrates that AI-driven personalized promotion engines represent a significant and accessible opportunity for supermarket retailers of all sizes to improve their marketing efficiency, customer engagement, and profitability. The proposed system provides a practical, deployable, and economically viable alternative to both generic broadcast promotions and expensive enterprise personalization platforms, with strong empirical support for its business value. As the retail sector continues to digitize and customer expectations for personalization continue to rise, systems of this kind will become an increasingly essential component of competitive supermarket operations.

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GLOSSARY

Term	Definition
Association Rule Mining	A data mining technique that discovers co-occurrence patterns in transaction data, expressed as if-then rules with support, confidence, and lift metrics.
Causal Inference	A statistical methodology for estimating the causal effect of an intervention (e.g., a promotion) on an outcome (e.g., purchase), distinct from correlation-based prediction.
Collaborative Filtering (CF)	A recommendation technique that predicts user preferences based on the preferences of similar users, without requiring explicit product attribute information.
Content-Based Filtering (CBF)	A recommendation technique that suggests items similar to those a user has previously interacted with, based on item attribute profiles.
Click-Through Rate (CTR)	The ratio of users who click on a promotion to the total number of users who view the promotion, expressed as a percentage.
Cold Start	The challenge faced by recommendation systems when insufficient historical data is available for new users or new items.
Collaborative Filtering	See CF above.
Customer Lifetime Value	The total revenue a business can expect from a customer

(CLV)	over the entire duration of the customer relationship.
FP-Growth	An efficient algorithm for association rule mining that avoids the computationally expensive candidate generation step of the Apriori algorithm.
K-Means Clustering	An unsupervised learning algorithm that partitions data into k clusters by minimizing within-cluster variance.
Lift	An association rule metric measuring how much more likely two items are to co-occur together than by chance alone.
Precision@K	A recommendation evaluation metric measuring the proportion of relevant items among the top-K recommendations.
Random Forest	An ensemble machine learning algorithm that builds multiple decision trees and combines their predictions to improve accuracy and reduce overfitting.
RFM Analysis	A customer segmentation framework based on Recency, Frequency, and Monetary value of customer transactions.
ROI (Return on Investment)	A performance metric measuring the efficiency of an investment, calculated as (Gain - Cost) / Cost × 100%.
Silhouette Score	A clustering quality metric ranging from -1 to 1 that measures how similar each data point is to its own cluster

	compared to other clusters.
SVD (Singular Value Decomposition)	A matrix factorization technique used in collaborative filtering to decompose the user-item interaction matrix into lower-dimensional latent factor representations.
Uplift Modeling	A predictive modeling technique that estimates the incremental effect of an action (treatment) on an outcome, distinguishing between customers who would respond to an incentive and those who would not.
XAI (Explainable AI)	A set of techniques and frameworks that make AI model decisions interpretable and understandable to human users.

APPENDICES

Appendix A: Sample Campaign Output File Structure

The following describes the structure of the campaign output CSV files generated by the promotion engine for each product campaign. These files are generated automatically and stored in the campaign_outputs/ directory.

File naming convention: campaign_{ProductID}_{DiscountPct}pct_{YYYYMMDD}_{HHMMSS}.csv

File contents:

- CustomerID - Unique customer identifier
- Name - Customer full name
- Age - Customer age
- Gender - Customer gender
- Location - Customer location/region
- CustomerSegment - Cluster assignment (frequent/regular/occasional/at-risk)
- PurchaseProbability - Predicted purchase probability from RF classifier (0.0-1.0)
- UpliftScore - Estimated incremental purchase probability due to promotion
- ProductID - Promoted product identifier
- ProductName - Promoted product name
- ProductCategory -Product category (e.g., Bakery, Dairy)
- DiscountPercentage - Applied discount (e.g., 10%)
- OriginalPrice - Product original price (LKR)
- DiscountedPrice - Promotion price (LKR)
- CampaignID - Unique campaign identifier
- GeneratedAt - Timestamp of campaign generation